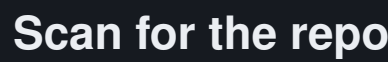


Electrical Control & Power System

Portland State University • MCECS • Department of Electrical & Computer Engineering

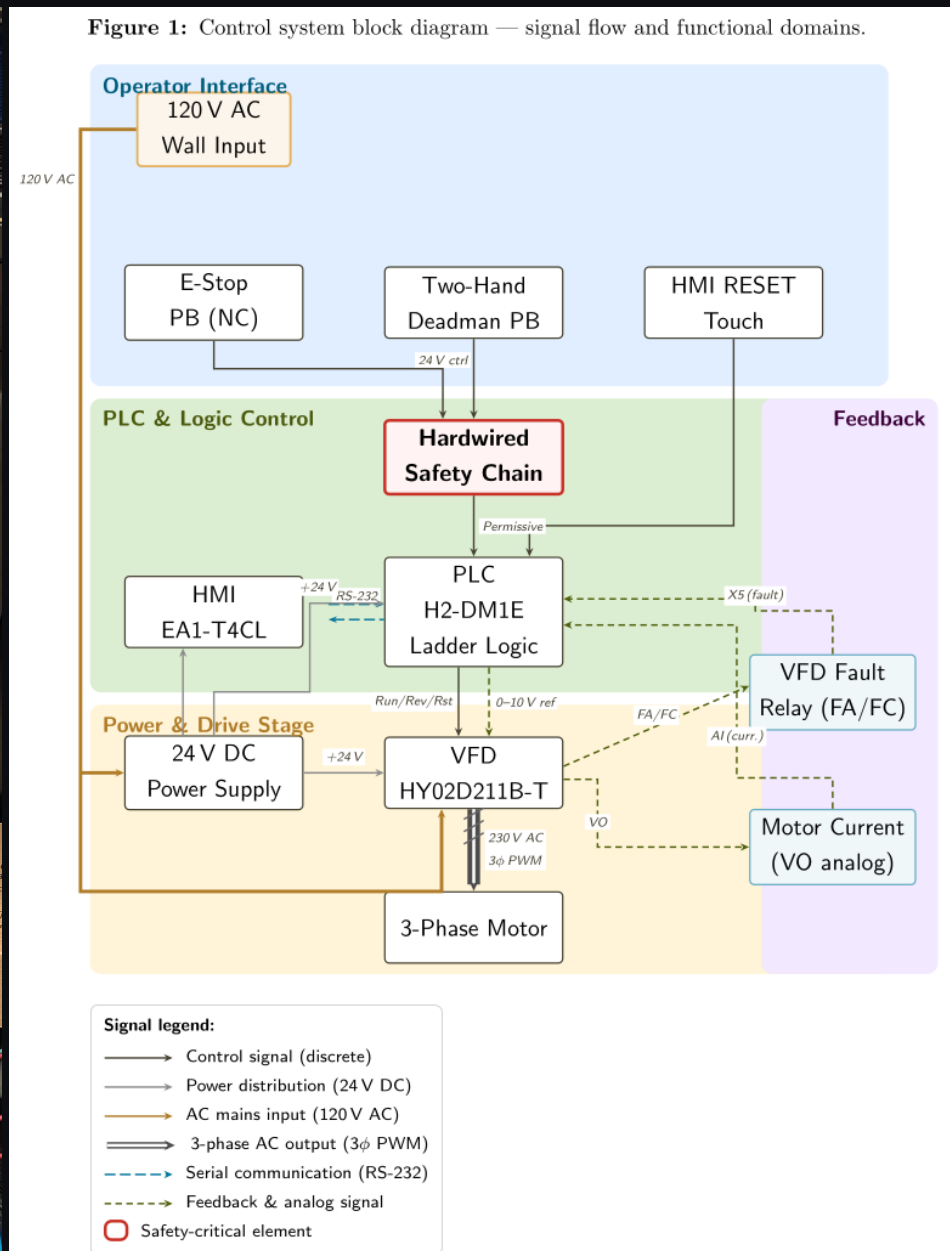


- Ichor Systems creates large volumes of thick rigid plastic waste (PVC, PP, LDPE) from semiconductor manufacturing.
- Sheets are 0.25–0.5 in. thick, 4–12 in. wide – off-the-shelf shredders are too small or too expensive.
- Our scope: **electrical control + power system**. Mechanical frame and blades come from the ME team.

- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

- **Industrial stack:** AutomationDirect end-to-end. No hobby boards in the control or safety path.
- **Power path:** PowerTec 71008 disconnect → Mean Well 24VDC PSU → SD-N35 contactor → Huanyang VFD → motor.
- **Motor:** IronHorse MTCP2-002, 2 HP, 230 V, 1735 RPM, VFD-rated.
- **Safety:** E-stop hardwired through the contactor coil, parallel to the PLC.



Operator face • panel interior • system block diagram.

- **CPU:** DirectLOGIC 205 + H2-DM1E.
- **I/O:** 8 DC in, 8 relay out, 8 analog in, 4 RTD in, 0–10 V analog out to the VFD.
- **Program:** 31 rungs in Do-more. Deterministic state machine, 3-strike jam recovery, hard-lockout path.

- **Project:** 4 screens, 27 PLC tags, serial to the CPU.



- **Power:** panel 24 VDC → LM2596 buck → 3.3 V to ESP32.



Status: bench-validated. Full-machine test pending the ME prototype.



150% over-torque threshold on the VFD.

Contactor: Mitsubishi SD-N35.

control-panel budget, so the analog V_O/V_I path was used instead).

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